



▲ TROPICAL SEAS

The world's oceans include a variety of environments. At one extreme, the warm, clear surface waters of the tropics support a dazzling variety of life, but often in small numbers because the clear waters contain very little food.



▲ ICY OCEANS

At the other extreme, surface waters near the poles freeze over for much of the year. Only a few types of animals can cope with this, but they can form huge populations because the cool, cloudy waters are rich in food resources.



THE WORLD'S OCEANS



The world's oceans are separated by the continents into five main oceans, with the Atlantic, Pacific, and Indian Ocean linked in the south by the Southern Ocean. Each ocean has its own character, but water constantly flows from one to another.

SIZE AND DEPTH (M=million)

OCEAN	AREA	AVERAGE DEPTH
Pacific Ocean	64 M sq miles (166 M sq km)	14,042 ft (4,280 m)
Atlantic Ocean	32 M sq miles (82 M sq km)	10,827 ft (3,300 m)
Indian Ocean	29 M sq miles (74 M sq km)	12,763 ft (3,890 m)
Arctic Ocean	5 M sq miles (12 M sq km)	3,248 ft (990 m)
Southern Ocean	14 M sq miles (35 M sq km)	10,991 ft (3,350 m)

▲ THE WATER OF LIFE

Liquid water is essential to life on Earth, and probably anywhere else in the universe. It is a vital part of the tissues of living things, and it carries the chemicals that form the complex molecules of proteins and deoxyribonucleic acid (DNA). The presence of these chemicals in seawater may have enabled life to begin in the oceans more than 3.5 billion years ago.

OFFSHORE OIL RIG ►

The oceans are rich sources of food, oil, minerals, and other natural resources, and they have been exploited as trade routes for centuries. But they are also dangerous environments with immense destructive power and are impossible to explore without special equipment. This is why the oceans are the last frontier of exploration on Earth.

